

Supply Chain Analysis

1. Project Overview :-

Quick-commerce delivery companies (Jio Mart, Zepto, Blinkit, Big Basket, Amazon Now, Swiggy Instamart, Flipkart Minutes, Dunzo) na order, delivery, ane customer satisfaction data nu analysis kari, business insights provide karva mate interactive Power BI dashboard banavyu.

2. Dataset Summary :-

CSV file - customer table

- Rows: 1,00,000

- Columns: 13

Fields: Order_ID, Company, City, Customer_Age, Order_Value, Delivery_Time_Min, Distance_Km, Items_Count, Product_Category, Payment_Method, Customer_Rating, Discount_Applied, Delivery_Partner_Rating

3. Exploratory Data Analysis using Python :-

- **Data Loading:** Imported the dataset using *pandas*.
- **Initial Exploration:** Used *df.info()* to check structure and *.describe()* for summary statistics.

	Order_ID	Customer_Age	Order_Value	Delivery_Time_Min	Distance_Km	Items_Count	Customer_Rating	Discount_Applied	Delivery_Partner_Rating
count	1.000000e+06	1000000.000000	1000000.000000	1000000.000000	1000000.000000	965000.000000	953000.000000	1000000.000000	895863.000000
mean	1.500000e+06	38.470387	571.643584	16.452412	7.750231	9.998317	3.042072	0.400883	3.749257
std	2.886753e+05	12.118520	382.784520	6.257172	4.186665	5.478725	1.186872	0.490078	0.722121
min	1.000001e+06	18.000000	50.000000	5.000000	0.500000	1.000000	1.000000	0.000000	2.500000
25%	1.250001e+06	28.000000	287.840150	12.368000	4.130000	5.000000	2.000000	0.000000	3.100000
50%	1.500000e+06	38.000000	531.324600	16.342000	7.740000	10.000000	3.000000	0.000000	3.700000
75%	1.750000e+06	49.000000	796.182500	20.196000	11.380000	15.000000	4.000000	1.000000	4.400000
max	2.000000e+06	59.000000	13877.325000	40.000000	15.000000	19.000000	5.000000	1.000000	5.000000

- **Missing Data Handling:** Checked for null values and imputed missing values in the *City*, *Items_Count*, *Customer_Rating*, *Delivery_Partner_Rating* columns using the median rating of each product category.

```

Order_ID          0
Company           0
City              0
Customer_Age      0
Order_Value       0
Delivery_Time_Min 0
Distance_Km       0
Items_Count       0
Product_Category  0
Payment_Method     0
Customer_Rating   0
Discount_Applied  0
Delivery_Partner_Rating 0
dtype: int64

```

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions):-

1. **Revenue by City –** Compared total revenue (Order_Value) generated across different cities.

Data Output

Messages

Notifications

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SQL

	City	total_revenue	
	text	numeric	
1	Gurgaon	54963765.81	
2	Noida	54019300.26	
3	Delhi	48004056.68	
4	Mumbai	46842186.77	
5	Bengluru	46142373.89	
6	Amritsar	44646384.48	
7	Hyderab...	44646205.93	
8	Kolkata	44488389.49	
9	Chennai	44234172.13	
Total rows: 13		Query complete 00:00:00.390	

2. **High-Delay Discount Orders –** Identified orders that used a discount but still had delivery time greater than the average delivery time.

Data Output Messages Notifications			
	Order_ID bigint	Delivery_Time_Min double precision	Order_Value double precision
1	1000001	19.182	702.3375
2	1000017	17.386	50
3	1000022	17.804	716.8986
4	1000024	19.592	934.74
5	1000025	20.01	1147.9395
6	1000027	20.274	690.88425
7	1000029	34.382	516.62175
8	1000037	25.988	576.567
9	1000038	17.03	851.2365
Total rows: 197238		Query complete 00:00:00.869	

3. **Top 5 Categories by Rating** – Found product categories with the highest average customer ratings.

Data Output Messages Notifications		
	Product_Category text	avg_customer_rating numeric
1	Fruits & Vegetables	3.04
2	Personal Care	3.04
3	Dairy	3.04
4	Beverages	3.04
5	Groceries	3.04

4. **Discount Impact on Delivery Time** – Compared average delivery time between orders with and without discount applied.

Data Output Messages Notifications		
	Discount_Applied bigint	avg_delivery_time numeric
1	0	16.45
2	1	16.45

5. **High vs Low Rated Delivery Partners** – Compared average delivery time and SLA breach % between partners rated 4+ and below 4.

Data Output		Messages	Notifications	
				
			SQL	
	partner_rating_group text	total_orders bigint	avg_delivery_time numeric	sla_breach_pct numeric
1	High Rated (4+)	375754	16.44	59.26
2	Low Rated (<4)	624246	16.46	59.20

6. **Delay-Prone Categories** – Identified 5 product categories with the highest percentage of delayed orders (delivery time > 15 min).

Data Output

Messages

Notifications

SQL

	Product_Category text	delay_rate numeric
1	Personal Care	59.35
2	Household	59.29
3	Groceries	59.26
4	Dairy	59.23
5	Fruits & Vegetables	59.19

7. **Distance-Based Segmentation** – Classified orders into Near, Medium, and Far segments based on distance, and compared average delivery time.

SELECT * FROM

Data Output

Messages

Notifications

distance_segment

text

number_of_orders

bigint

avg_delivery_time

numeric

1

Far

689366

17.74

2

Medium

206933

14.00

3

Near

103701

12.76

8. **Top 3 Categories per City** – Listed the most ordered product categories within each city.

	category_rank bigint	City text	Product_Category text	total_orders bigint
1	1	Amritsar	Groceries	11344
2	2	Amritsar	Personal Care	11316
3	3	Amritsar	Dairy	11308
4	1	Bengluru	Beverages	11433
5	2	Bengluru	Groceries	11386
6	3	Bengluru	Dairy	11373
7	1	Chennai	Groceries	11425
8	2	Chennai	Beverages	11357
9	3	Chennai	Snacks	11319
Total rows: 39		Query complete 00:00:00.470		

9. **Customer Rating vs Delivery Performance** – Checked whether orders with low customer ratings (<3) had longer delivery times and greater distances.

	rating_group text	total_orders bigint	avg_delivery_time numeric	avg_distance numeric
1	Good Rating (3...	545779	16.30	7.75
2	Low Rating (<3)	454221	16.63	7.75

10. **SLA Breach by City** – Calculated the percentage of delayed orders (SLA breach) contributed by each city.

Data Output Messages Notifications				
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	City text	total_orders bigint	delayed_orders bigint	sla_breach_pct numeric
1	Haridwar	78958	78753	99.74
2	Jaipur	78901	69671	88.30
3	Mumbai	79078	56182	71.05
4	Bengluru	78973	50041	63.36
5	[null]	52000	29051	55.87
6	Pune	78580	43765	55.69
7	Amritsar	79074	44005	55.65
8	Gurgaon	79004	43961	55.64
9	Chennai	78725	43715	55.53
Total rows: 13		Query complete 00:00:00.454		

11. **Company-wise SLA Performance** – Compared average delivery time and SLA breach % across different delivery companies.

Data Output Messages Notifications				
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	Company text	total_orders bigint	avg_delivery_time numeric	sla_breach_pct numeric
1	Jio Mart	124783	22.96	93.48
2	Amazon Now	124658	18.94	82.54
3	Big Basket	124471	17.98	75.19
4	Flipkart Minutes	125542	16.99	66.71
5	Swiggy Instam...	125119	16.02	57.80
6	Blinkit	125464	15.07	48.69
7	Dunzo	124900	14.12	39.34
8	Zepto	125063	9.57	10.22
Total rows: 8		Query complete 00:00:00.303		

12. **Payment Method vs Order Value** – Identified which payment method is associated with the highest average order value.

1.1.7 SELECT

Data Output Messages Notifications

	Payment_Method text	avg_order_value numeric	total_orders bigint
1	Cash on Delivery	573.42	200739
2	UPI	571.92	199882
3	Debit Card	571.16	200038
4	Wallet	571.11	199457
5	Credit Card	570.60	199884

13. **Age Group vs Delivery Delays –** Analyzed whether customer age groups affect delivery time and SLA breach percentage.

Data Output Messages Notifications

	age_group text	total_orders bigint	avg_delivery_time numeric	sla_breach_pct numeric
1	18-24	167064	16.44	59.11
2	25-40	381644	16.45	59.19
3	41-60	451292	16.46	59.28

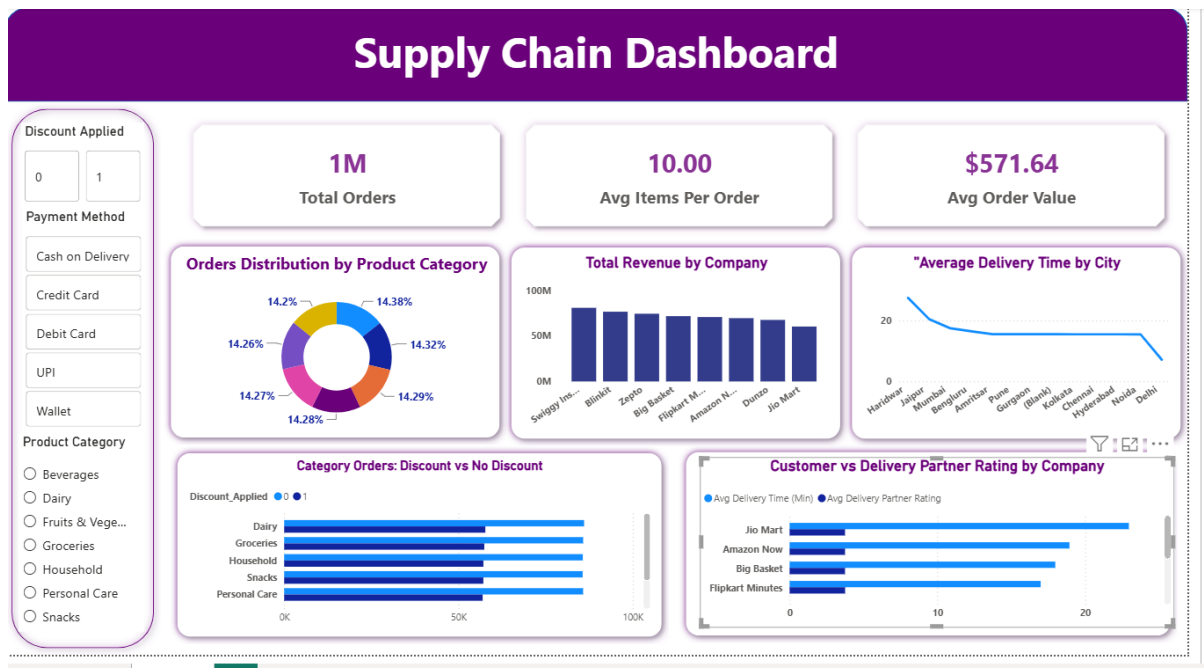
14. **Item Count vs Delivery Time –** Checked whether the number of items in an order affects delivery time.

Data Output Messages Notifications

	item_count_group text	avg_delivery_time numeric	total_orders bigint
1	Low (1-3 items)	16.46	152447
2	High (8+ items)	16.45	644241
3	Medium (4-7 items)	16.45	203312

5. Dashboard in Power BI :-

Finally, we built an interactive dashboard in Power BI.



6. Business Recommendations :-

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.